The Mechanism of Thermalization in the Bost–Connes System

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Abstract

The Bost–Connes system supplies KMS equilibrium states, but the existing evolution generated by the evaluation layer contains no dynamics that carries a general initial state to equilibrium: the evaluation algebra and its inner fluctuations preserve every valuation shell, and the cross-shell probability flux vanishes. This paper constructs the missing dynamics. The structure supplied is the thermalization dilation, consisting of the adjacent-shell shift, its minimal unitary dilation, and the Fourier unitary realizing the finite Weyl conjugation; its physical content is a chain of repeated collisions: environment modes exchange single quanta with the shell levels under energy conservation, and the weak-coupling limit converges to a unique GKSL generator whose jump-rate ratio is locked by the KMS condition to $p^{-\beta}$. The environment carrier passes a threefold screening by conservation laws, energy, and records, and under the particle identification of the equilibrium framework is uniquely determined to be the photon. Once the mechanism is in place, the arrow of time is a theorem rather than an assumption: at fixed temperature the relative entropy decreases monotonically toward equilibrium, the mean entropy production, the trajectory-wise fluctuations, and the escaping records all point the same way, and the entire time asymmetry is confined to a single boundary condition, environment refreshing. In units of the total rate, the relaxation spectrum is completely determined by tree geometry and detailed balance and contains no adjustable parameter: the infinite-depth spectral gap $(\sqrt{p}-1)^2/(p+1)$ attains the bound permitted by regular-tree geometry, the characteristic polynomial is independent of all microscopic branch weights under the constraints of tree structure, fixed escape-rate profile, and no holding at the zero element, the channels relax in layers indexed by the conductor of the multiplicative character, and the slowest mixing—that of the generation labels—obeys an Arrhenius law with activation energy equal to the conductor depth times $\log p$ (referred to the unit-shell damping). The Fourier realization and the minimal unitary dilation are unique up to a phase and up to unitary equivalence on the auxiliary space, respectively; the Fourier eigenphase of the real quadratic character is fixed by a Gauss sum, providing the test quantities for the Majorana phase criterion and for controlled-collision experiments.

Contents

1 Shell-occupation conservation

The KMS condition fixes the equilibrium distribution but supplies no path along which a general initial state approaches it. For a fixed prime p, if an initial state and the Gibbs state have different valuation-shell occupations, thermalization requires at least one nonzero inter-shell probability flux. Whether the existing dynamics can thermalize therefore hinges on whether it can change the expectation values of the shell projections; if all generators preserve these projections, a nonequilibrium shell distribution can never relax to the Gibbs distribution.

1.1 Shell decomposition and Gibbs equilibrium

A local state carries both the degenerate degrees of freedom inside a single valuation shell and the occupation distribution across different shells. Shell-occupation conservation concerns only the latter, so we first reduce the KMS equilibrium state to the valuation shells and identify the probability variables that thermalization must change.

Fix an odd prime p and a truncation depth K; the local state space is

$$\mathcal{H}_{p,K} = L^2(\mathbb{Z}/p^K\mathbb{Z}), \tag{1}$$

with the residue classes $|x\rangle$ ($x \in \mathbb{Z}/p^K\mathbb{Z}$) as the position basis. The p-adic valuation $v(x)$ is the largest integer j with $p^j \mid x$, with the convention $v(0) = K$. The nonzero elements split into shells by valuation:

$$S_j = \{x : v(x) = j\}, \qquad |S_j| = (p-1)p^{K-j-1}, \qquad 0 \le j < K, \tag{2}$$

and the zero element forms the terminal shell $S_K = \{0\}$. Write P_j for the projection onto S_j. For $j < K$ the shell-average state is

$$e_j = |S_j|^{-1/2} \sum_{x \in S_j} |x\rangle, \qquad e_K = |0\rangle, \tag{3}$$

and the shell-average subspace is

$$\mathcal{H}^{\text{shell}}_{p,K} = \operatorname{span}\{e_0, \ldots, e_K\}. \tag{4}$$

The Hamiltonian of the valuation direction acts on this subspace as

$$H = (\log p)N, \qquad Ne_j = je_j. \tag{5}$$

The energy difference between adjacent shells is always $\log p$. At inverse temperature β, the Gibbs weights of the shell-reduced state are

$$\pi_j(\beta) = \frac{(1-p^{-\beta})p^{-\beta j}}{1-p^{-\beta(K+1)}}, \qquad 0 \le j \le K. \tag{6}$$

These weights agree with the shell reduction of the BC KMS$_\beta$ state: $\varphi_\beta(\mu_{p^j}\mu^*_{p^j}) = p^{-j\beta}$, which after truncated normalization is precisely (6). This distribution is the Gibbs distribution of H on the shell-average subspace; if H is extended to the full position space, the intra-shell degeneracies must be counted as well. The Haar shell masses of the finite ring are

$$h_j = (1-p^{-1})p^{-j} \quad (j < K), \qquad h_K = p^{-K}. \tag{7}$$

At $\beta = 1$ and $K \to \infty$ one has $\pi_j(1) \to h_j$ for each fixed j; at finite K the terminal shell retains a boundary correction. The global Bost–Connes partition function $\zeta(\beta) = \prod_q (1-q^{-\beta})^{-1}$ has a pole at $\beta = 1$[1, 3], while the local factor at fixed p has no phase transition; the critical readout later in the paper is selected by the global structure (for the standard KMS background see [2]).

For an arbitrary state ρ_t, the j-th shell occupation is

$$q_j(t) = \operatorname{Tr}(P_j\rho_t). \tag{8}$$

If $q_j(0) \neq \pi_j(\beta)$, relaxation to Gibbs equilibrium requires the shell occupations to change in time. When the shell occupations admit a closed Markov description, the continuity equation reads

$$\dot{q}_j(t) = \sum_{k \neq j} \big(J_{k\to j}(t) - J_{j\to k}(t)\big), \tag{9}$$

where $J_{j\to k}$ are the inter-shell probability fluxes. Whether the existing dynamics can thermalize thus reduces to an operator question: whether it can produce nonzero cross-shell matrix elements.

1.2 Shell invariance of the evaluation algebra

We now check whether cross-shell matrix elements exist in the existing evaluation algebra. Position functions, character projections, and conditional expectations are intra-shell readouts; unit scalings, the involution, complex conjugation, and the convolution operator are intra-shell propagation. The inter-shell behavior of both kinds is governed by whether they preserve the p-adic valuation.

A position function $f(x)$ is diagonal in the position basis and hence commutes with every P_j. The unit scaling

$$M_u|x\rangle = |ux\rangle, \qquad u \in (\mathbb{Z}/p^K\mathbb{Z})^\times, \tag{10}$$

satisfies $v(ux) = v(x)$; the involution $\iota|x\rangle = |-x\rangle$ satisfies $v(-x) = v(x)$, and complex conjugation also preserves the position basis. The convolution operator invokes only invertible multiplications with $p \nmid n$, so that

$$v(nx) = v(x) \qquad (p \nmid n). \tag{11}$$

Its multiplicative-character decomposition proceeds shell by shell. Multiplicative-character projections are weighted averages of unit scalings and preserve shells because each M_u does; the conditional expectation onto the shell-preserving subalgebra likewise produces no cross-shell matrix element. Appendix A verifies every evaluation component item by item.

Write $\mathcal{A}_{\text{eval}}$ for the $*$-algebra generated by the operators above (complex conjugation is an antilinear map, assigned to the real structure of the next paragraph; the conditional expectation is a shell-preserving map on operators whose range lies inside this algebra, and is not listed as a generator). All generators preserve the shell decomposition, and sums, products, adjoints, and norm closure do not change this property, so

$$\boxed{[A, P_j] = 0, \qquad A \in \mathcal{A}_{\text{eval}}, \quad 0 \le j \le K.} \tag{12}$$

Gauge inner fluctuations do not open cross-shell directions either. The finite Dirac-type operator of the equilibrium framework acts along the unit-group directions (for the construction see [30]); this paper uses only one of its properties, declared as an input condition: $[D, P_j] = 0$. For arbitrary $a_r, b_r \in \mathcal{A}_{\text{eval}}$, the one-form

$$A_D = \sum_r a_r[D, b_r] \tag{13}$$

still commutes with all P_j. The real structure is given by the involution together with complex conjugation and preserves the valuation, so $A_D + JA_DJ^{-1}$ is also shell block-diagonal. The existing evaluation algebra and its inner fluctuations can only change intra-shell structure.

1.3 Shell-occupation conservation

That each operator preserves the valuation does not yet rule out that their combinations in the Hamiltonian, the jump terms, and the inner fluctuations produce inter-shell probability flux. The generators of the physical dynamics are taken from the evaluation layer—an input clause of the equilibrium framework[30] (the p-direction isometry of the BC algebra itself crosses shells and is not included; see the Appendix). Let $\mathcal{G}$ be a GKSL generator[5, 6] whose Hamiltonian part H' and jump operators L_α are taken from $\mathcal{A}_{\text{eval}}$ or its inner-fluctuation one-forms (both commute with all P_j). By (12),

$$[H', P_j] = [L_\alpha, P_j] = 0. \tag{14}$$

Applying the dual generator to a shell projection gives

$$\mathcal{G}^*(P_j) = i[H', P_j] + \sum_\alpha \left(L_\alpha^\dagger P_j L_\alpha - \frac{1}{2}\{L_\alpha^\dagger L_\alpha, P_j\} \right) = 0. \tag{15}$$

Therefore

$$\frac{d}{dt} q_j(t) = \frac{d}{dt} \operatorname{Tr}(P_j \rho_t) = \operatorname{Tr}(\mathcal{G}^*(P_j)\rho_t) = 0 \tag{16}$$

holds for all j.

$$\boxed{\text{The dynamics generated by the existing evaluation algebra and its inner fluctuations preserves every valuation-shell occupation.}} \tag{17}$$

If the initial state satisfies $q(0) \neq \pi(\beta)$, equation (16) rules out its relaxation to the shell Gibbs distribution. The existing dissipation can still copy shell labels and suppress inter-shell coherence, and it can rearrange states inside each shell; these processes leave the q_j unchanged and are therefore not a thermalization of the shell occupations.

Changing the shell occupations requires nonzero cross-shell matrix elements. The adjacent-shell shift and its unitary realization thus become the starting point of the thermalization dilation.

2 The thermalization dilation

Changing the shell occupations requires cross-shell motion, but an arbitrary cross-shell matrix does not yet constitute physical dynamics. The minimal transport should connect only adjacent shells with energy difference $\log p$, admit a unitary realization on a larger space, and preserve the original Weyl commutation structure of the finite state space. The p-direction of the Bost–Connes multiplicative semigroup supplies the change of valuation; the Hecke adjacency supplies its microscopic branching; and the Fourier unitary pins down the conjugation structure of the finite phase space.

2.1 The adjacent-shell shift

The energy difference between adjacent shells is $\log p$. The minimal cross-shell motion should carry S_j to S_{j+1} while keeping the intra-shell labels. The p-direction of the multiplicative semigroup has this action; the many-to-one nature of the finite truncation then requires extracting its normalized shell-average action.

The state space decomposes by valuation as

$$\mathcal{H}_{p,K} = \bigoplus_{j=0}^{K} \mathcal{H}_j, \qquad \mathcal{H}_j = P_j \mathcal{H}_{p,K}. \tag{18}$$

Equation (12) shows that all existing evaluation operations are block-diagonal with respect to this decomposition. If the shell occupations of an initial state differ from the Gibbs weights, intra-block motion can only rearrange the states inside each $\mathcal{H}_j$; it cannot produce a flux that transports probability from $\mathcal{H}_j$ to $\mathcal{H}_{j+1}$. The first input required for thermalization is therefore a cross-shell matrix element. Adjacency follows from the single-frequency structure of the environment: if one cross-shell step exchanged $m \geq 2$ energy quanta (frequency $m \log p$), the shell index would be conserved modulo m step by step, the chain would lose irreducibility on the shell ladder and could no longer carry an arbitrary initial state to the Gibbs distribution; in a single-frequency $\log p$ environment the cross-shell motion must therefore contain the one-step channel, and higher multi-quantum channels are not introduced, by the lowest-order (single-quantum exchange) coupling convention.

The Bost–Connes representation contains the p-direction isometry μ_p of the multiplicative semigroup. It raises the valuation by one step; its truncated form on the finite ring satisfies

$$v(px) = v(x) + 1 \tag{19}$$

(before the terminal shell is reached). But $x \mapsto px \bmod p^K$ is a p:1 map: the p inputs differing by p^{K-1} land on the same output. It discards the top digit and cannot serve directly as the unitary time evolution of the finite system. On the other hand, μ_p alone gives only a directed isometry; it contains neither the reverse jumps nor the heat-bath weights, and is not yet a thermalization mechanism.

The required system-side object acts on the shell-average subspace $\mathcal{H}_{p,K}^{\text{shell}} = \text{span}\{e_0, \dots, e_K\}$ (equation (4)). Take the finite chain shift

$$Ve_j = e_{j+1} \quad (0 \le j < K), \qquad Ve_K = 0, \tag{20}$$

whose adjoint satisfies

$$V^\dagger e_{j+1} = e_j, \qquad V^\dagger e_0 = 0. \tag{21}$$

V is the normalized model of μ_p on the shell-average subspace. The finite-ring map $|x\rangle \mapsto |px\rangle$ carries a normalization factor $\sqrt{p}$ on the interior shells, with a different factor at the terminal boundary. V and $V^\dagger$ provide the raising and lowering operators for absorbing and emitting one energy quantum $\log p$.

2.2 The minimal unitary dilation

The shift operator V supplies the system-side raising motion, but the terminal shell makes it a partial isometry. A closed-system evolution preserves norms, so V can only be the compression of a larger unitary evolution to the system subspace.

Embed $\mathcal{H}_{p,K}^{\text{shell}}$ isometrically into $\mathcal{K} = \ell^2(\mathbb{Z})$:

$$Ie_j = \delta_j, \qquad 0 \le j \le K, \tag{22}$$

and take the bilateral shift on $\mathcal{K}$:

$$U_V \delta_n = \delta_{n+1}, \qquad n \in \mathbb{Z}. \tag{23}$$

U_V is unitary and satisfies, for every $m \ge 0$,

$$I^* U_V^m I = V^m. \tag{24}$$

(Basis-wise verification: $I^* U_V^m I e_j$ equals e_{j+m} for $j+m \le K$ and 0 otherwise, matching $V^m e_j$.) Moreover,

$$\mathcal{K} = \overline{\text{span}}\{U_V^n I\psi : \psi \in \mathcal{H}_{p,K}^{\text{shell}},\ n \in \mathbb{Z}\}, \tag{25}$$

so U_V is the minimal unitary dilation of V[29] (the spanning condition is verified directly: the vectors $U_V^n I e_0$ run through the standard basis of $\ell^2(\mathbb{Z})$). The finite shell shift on the system side is recovered by (24), and the components that cross the terminal boundary are stored in the dilation space.

While the dilation degrees of freedom are retained, the process described by U_V is reversible. Only after $V, V^\dagger$ are placed in an energy-conserving system–environment coupling, with the outgoing modes departing and being traced out, does one obtain irreversible reduced dynamics.

2.3 The Hecke adjacency structure

The shell-average shift retains the change of valuation but erases the microscopic branches contained in one and the same jump. The number of branches gives the microscopic degeneracy of a cross-shell jump, and at the critical temperature its equal-weight count coincides with the environment rate ratio (§3.2); one must therefore return to the adjacency structure of p-adic lattices. With $K_0 = \mathrm{GL}_2(\mathbb{Z}_p)$ as the vertex stabilizer (up to center), the Hecke double coset that changes the lattice index by one step has the standard right-coset decomposition[22]

$$K_0 \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} K_0 = \bigsqcup_{b \in \mathbb{F}_p} \begin{pmatrix} p & b \\ 0 & 1 \end{pmatrix} K_0 \sqcup \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} K_0. \tag{26}$$

These matrices act on homothety classes of p-adic lattices, or on $\mathbb{P}^1(\mathbb{Q}_p)$; their domain of action is distinguished from the finite ring $\mathbb{Z}/p^K\mathbb{Z}$. Equation (26) yields the $p+1$ neighbors of each vertex of the Bruhat–Tits tree.

After choosing an end of the tree and taking the Busemann coordinate, the $p+1$ edges aggregate into two classes: one edge raises the valuation coordinate by one step, and p edges lower it by one step. V and $V^\dagger$ are the normalized interfaces of these two motions in the shell-average coordinate; the p descending edges aggregate into the degeneracy of the descending channel.

The rates are fixed solely by the environment; the directional degeneracy enters only the internal apportionment of the descending step (its concrete form is the position kernel of §4.1). The coincidence of the equal-weight count with the detailed-balance ratio at the critical temperature (§3.2) is a consistency check, not an independent factor multiplying the rates. The Hecke adjacency acts on the tree, while the finite Fourier transform acts on the finite Heisenberg phase space.

2.4 Finite Weyl compatibility

The Hecke adjacency fixes the branching structure on the tree but does not act on the position phases of the finite state space. The position phases and the additive translations generate the finite Heisenberg group; if the dilation is to retain the original observable algebra, it must maintain the Weyl commutation relations between the two.

For $r, c \in \mathbb{Z}/p^K\mathbb{Z}$, the position phase and the additive translation are

$$\varepsilon_r|x\rangle = e^{2\pi i r x/p^K}|x\rangle, \qquad T_c|x\rangle = |x+c\rangle. \tag{27}$$

Together they generate the finite Heisenberg group. The discrete Fourier transform

$$F|x\rangle = p^{-K/2} \sum_y e^{2\pi i x y/p^K}|y\rangle \tag{28}$$

satisfies

$$F\varepsilon_r F^{-1} = T_{-r}, \qquad FT_c F^{-1} = \varepsilon_c, \qquad F^2 = \iota. \tag{29}$$

Thus F exchanges the position-phase and translation directions while preserving the central character of the Heisenberg group. Quadratic phases, unit scalings, and conjugation by F give the Weil representation of $\mathrm{SL}_2(\mathbb{Z}/p^K\mathbb{Z})$ (up to projective phases)[24, 13], so the commutation relations of the old operations survive in the dilated finite phase space.

At this point the Hecke adjacency on the tree and the Weyl conjugation on the finite phase space have their separate domains of action; the adjacent-shell shift, the unitary dilation, and the Fourier realization can be assembled into one compatible data set.

2.5 The compatible structure of the thermalization dilation

The adjacent-shell shift, the minimal unitary dilation, and the finite Weyl realization act on system transport, joint evolution, and conjugation symmetry, respectively. The **thermalization dilation** of the equilibrium framework (the evaluation structure of [30], whose local form is given in §1.2) is given by the triple

$$\mathfrak{X} = (F, V, U_V) \tag{30}$$

subject to:

(i) V is the adjacent-shell shift of (20)–(21) (adjacency is forced by the single-frequency argument of §2.1; the specific normalization is a convention);

(ii) U_V is the minimal unitary dilation of V and satisfies (24);

(iii) F satisfies the exact Weyl relations (29);

(iv) $V, V^\dagger$ admit a family of cross-shell dynamics with the shell Gibbs distribution (6) as stationary state and rate ratio $p^{-\beta}$.

The Hecke double coset supplies the geometric degeneracy structure for clause (i) and is not an operator inside the triple. The compatibility of the three items rests on two verifiable joint facts: $FM_uF^{-1} = M_{u^{-1}}$ makes F preserve the radial (shell-average) subspace, compatible with the domain of V; and $F^2 = \iota \in \mathcal{A}_{\mathrm{eval}}$ hooks the newly introduced Fourier unitary back into the existing evaluation structure. Equations (28), (20)–(21), and (23) give a concrete realization

$$\mathfrak{X}_0 = (F, V, U_V). \tag{31}$$

The first three items come directly from the construction of these operators; the fourth can always be satisfied by birth–death dynamics for any ladder structure obeying clause (i), and is listed to name its physical realization, supplied by the collision process of Section 3: $V, V^\dagger$ exchange energy $\log p$ at each step, the environmental KMS condition gives the up/down rate ratio $p^{-\beta}$, and the shell Gibbs distribution is the unique stationary distribution of that shell chain.

The finite Heisenberg representation fixes the F that realizes the prescribed Weyl automorphism up to a phase; the minimal unitary dilation of a fixed V is fixed up to unitary equivalence on the auxiliary space. "Minimal" is to be read component by component: the shift is restricted to adjacent shells, the dilation is taken minimal, the Fourier unitary is fixed up to a phase; no global minimality criterion is imposed. The dynamical layer only needs the already-constructed $V, V^\dagger$, placed into an energy-conserving quantum exchange.

3 The thermalization mechanism

The thermalization dilation is still kinematics: it prescribes which cross-shell jumps may occur, but gives neither the jump rates nor the direction of time. The dynamics requires an energy-conserving system–environment exchange, environmental records that leave the local system, and repeated interactions obeying the KMS ratio. Together these turn the adjacent-shell shift into a process that relaxes to equilibrium.

3.1 The quantum exchange channel

An adjacent-shell jump exchanges energy $\log p$, so the environment must contain a mode of the same frequency and realize absorption and emission through a coupling that conserves total energy. After the environment departs, the partial trace should convert the exchange records into occupation transport and coherence decay.

Once the prime p is fixed, the valuation shells S_j form a finite ladder of levels with spacing

$$\omega_p = \log p. \tag{32}$$

The partial isometry constructed in Section 2 acts on the shell-average basis as

$$Ve_j = e_{j+1}, \qquad V^\dagger e_{j+1} = e_j, \qquad 0 \le j < K, \tag{33}$$

with the endpoint conventions $Ve_K = 0$, $V^\dagger e_0 = 0$. An environment quantum of frequency ω_p interacts with this ladder of levels; the process consists of the following six steps.

Step 1: prepare the incident environment quantum. Let B_p denote a two-level environment mode with ground and excited states $|0\rangle_B$, $|1\rangle_B$ and Hamiltonian

$$H_B = \omega_p |1\rangle_B\langle 1|. \tag{34}$$

When the environment is at inverse temperature β, the state of a single incident mode is

$$\eta_{\beta,p} = \frac{|0\rangle\langle 0| + p^{-\beta}|1\rangle\langle 1|}{1+p^{-\beta}}, \qquad n_{\beta,p} := \langle 1|\eta_{\beta,p}|1\rangle = \frac{p^{-\beta}}{1+p^{-\beta}}. \tag{35}$$

The environment carrier is left unspecified here; §3.3 will screen the candidates by conservation laws, energy, and records.

Step 2: a single-quantum exchange occurs. The energy-conserving coupling between the system and B_p is taken to be

$$H_{\rm int} = g\big(V \otimes \sigma^- + V^\dagger \otimes \sigma^+\big), \qquad [H_{\rm int}, H + H_B] = 0, \tag{36}$$

where $\sigma^- = |0\rangle_B\langle 1|$. The collision lasts a time τ; write $\theta = g\tau$. The joint unitary is $U_\tau = e^{-i\tau(H+H_B+H_{\rm int})}$; since $[H_{\rm int}, H + H_B] = 0$, the free part contributes only a common phase inside each resonant doublet, and the exchange part acts as a rotation (the common phase is omitted below):

$$|e_j\rangle|1\rangle_B \longmapsto \cos\theta\,|e_j\rangle|1\rangle_B - i\sin\theta\,|e_{j+1}\rangle|0\rangle_B, \tag{37}$$

$$|e_{j+1}\rangle|0\rangle_B \longmapsto \cos\theta\,|e_{j+1}\rangle|0\rangle_B - i\sin\theta\,|e_j\rangle|1\rangle_B. \tag{38}$$

The first line says that the system absorbs one quantum of energy ω_p and rises from S_j to S_{j+1}; the second says that it emits a quantum of the same frequency and descends from S_{j+1} to S_j. The two processes are time reversals of each other, and the joint evolution remains unitary.

Step 3: the environment departs and the reduced channel forms. After the collision, the outgoing mode no longer interacts with the local system. The one-step state of the system is

$$\Phi_{\beta,\tau}(\rho) = \mathrm{Tr}_B\left[U_\tau(\rho \otimes \eta_{\beta,p})U_\tau^\dagger\right]. \tag{39}$$

Let $s^2 = \sin^2\theta$; the one-collision raising and lowering probabilities are

$$r_\uparrow = s^2 n_{\beta,p}, \qquad r_\downarrow = s^2(1-n_{\beta,p}), \qquad \frac{r_\uparrow}{r_\downarrow} = p^{-\beta}. \tag{40}$$

For the shell occupations $q_j = \langle e_j|\rho|e_j\rangle$, the one-step change of the interior levels is

$$q_j' - q_j = r_\uparrow q_{j-1} + r_\downarrow q_{j+1} - (r_\uparrow + r_\downarrow)q_j, \qquad 1 \le j \le K-1, \tag{41}$$

with nonexistent currents deleted at the endpoints and reflecting boundaries imposed. The local probability flux

$$J_{j\to j+1} = r_\uparrow q_j - r_\downarrow q_{j+1} \tag{42}$$

displays the occupation transport of thermalization directly: whenever adjacent shells fail the equilibrium ratio, a net flux exists; when all local fluxes vanish,

$$\frac{q_{j+1}}{q_j} = \frac{r_\uparrow}{r_\downarrow} = p^{-\beta}, \tag{43}$$

which after normalization is precisely the Gibbs shell distribution (6).

Step 4: exchange records decohere the state. The mixed thermal state (35) can be purified by an auxiliary degree of freedom that does not participate in the interaction; after the collision, "no exchange," "absorption," and "emission" correspond to distinct joint records of the auxiliary degree of freedom and the outgoing mode. When the environmental records are traced out, cross terms between different records do not enter (39). For a single collision, the jump Kraus operators of the channel are exactly

$$K_+ = -i\sqrt{r_\uparrow}\,V, \qquad K_- = -i\sqrt{r_\downarrow}\,V^\dagger, \tag{44}$$

marking the absorption and emission histories; the remaining Kraus operators give the no-exchange branch and guarantee trace preservation. A single collision need not remove all inter-shell coherence, because the equal level spacing allows different starting shells to exchange quanta of the same frequency; but the successive environmental output stores the entire exchange history. Under repeated partial traces, the interference between different histories is deleted step by step. On the shell-average subspace, the operator algebra generated by V and $V^\dagger$ is irreducible: only scalars commute with both. The jump operators generate the full algebra and the stationary state is faithful ($\pi_j > 0$), so the decoherence-free subalgebra is trivial[12]: no conserved sector carries inter-shell coherence. The full position space also contains intra-shell degrees of freedom; the reduced dynamics of their diagonal characters and of the coarse labels are computed in §4.2 and §4.3, respectively; this paper does not claim these three reduced readouts to be a complete decomposition of an arbitrary density matrix. The rearrangement of shell occupations and the decay of shell-average coherences are accomplished by one and the same exchange process, not by two separate mechanisms.

Step 5: environment refreshing yields irreversibility. If the collided mode returned, the system could reabsorb the quanta it emitted; the joint unitary evolution permits recurrence and cannot form a stable direction of time. Thermalization requires that after each collision the outgoing mode be moved into the environment's complement and a fresh mode in the same state (35) be sent in:

$$\rho_{m+1} = \Phi_{\beta,\tau}(\rho_m), \qquad \rho_m = \Phi^m_{\beta,\tau}(\rho_0). \tag{45}$$

Irreversibility does not come from U_τ; it comes from the outgoing records never returning to the local system, and from the local description tracing them out. Each step is still a unitary evolution on the joint space of the system and one fresh environment mode; the direction of the reduced semigroup is given by the one-way accumulation of environmental records. §3.4 will state where the records go: they are carried by the photon field (an additional environment outside the tensor-product structure, whose identity is fixed in §3.3); the role of the complement of the global prime tensor product is locational—records stored there commute with all p-local evaluation operators and persist to the end of the process.

Step 6: repeated collisions converge to the KMS state. In the weak-collision scaling $s^2 = \kappa\,\delta t$ (with $\delta t = \tau$), letting the number of collisions grow with $t = m\delta t$ fixed, (45) converges to

$$\frac{d\rho}{dt} = -i[H,\rho] + \Gamma_\uparrow \mathcal{D}[V]\rho + \Gamma_\downarrow \mathcal{D}[V^\dagger]\rho, \qquad \frac{\Gamma_\uparrow}{\Gamma_\downarrow} = p^{-\beta}, \tag{46}$$

where $\mathcal{D}[A]\rho = A\rho A^\dagger - \frac{1}{2}\{A^\dagger A, \rho\}$. $V, V^\dagger$ connect the entire finite shell ladder and $\Gamma_\uparrow, \Gamma_\downarrow > 0$, so the generator is irreducible on the shell-average subspace and the unique stationary state is the diagonal Gibbs state[11, 12]. Every initial state runs through

$$\boxed{\begin{array}{c}\text{incident environment quantum} \longrightarrow \text{resonant exchange} \longrightarrow \text{outgoing record} \\ \longrightarrow \text{partial trace} \longrightarrow \text{occupation transport and coherence decay} \longrightarrow \text{KMS equilibrium}\end{array}} \tag{47}$$

Equation (46) gives the form of the reduced dynamics (its pointwise form on position-diagonal distributions is given in §4.1), but the branching degeneracy in the rates and the environment carrier are still unspecified. The former is fixed by the adjacency geometry of the tree, the latter constrained by conservation laws and the record condition.

3.2 Branching geometry and detailed balance

At $\beta = 1$, the environmental KMS weights and the Hecke microscopic edge counts both give the up/down rate ratio 1:p. The two counts point to one and the same reduced jump and therefore must not be multiplied as independent factors.

The homothety classes of rank-two lattices form the $(p+1)$-regular Bruhat–Tits tree[22], with adjacency given by the $p+1$ representatives of (26). Choosing an end and taking the Busemann coordinate j (on the local model of §1.1, j is the valuation), each vertex has exactly one edge with $j \to j+1$ and p edges with $j \to j-1$. With equal edge weights the one-step probabilities are

$$P_\uparrow = \frac{1}{p+1}, \qquad P_\downarrow = \frac{p}{p+1}, \tag{48}$$

and the inter-shell detailed-balance ratio is

$$\frac{\pi_{j+1}}{\pi_j} = \frac{P_\uparrow}{P_\downarrow} = \frac{1}{p} = e^{-\log p}, \tag{49}$$

which is the same as the $\beta = 1$ Gibbs ratio of §1.1. The tree walk and the collision channel are the geometric and environmental descriptions of one and the same reduced jump, and are therefore not multiplied again. Equal weighting is guaranteed by the transitive action of $\mathrm{PGL}_2(\mathbb{Z}_p)$ on the $p+1$ directions; this belongs to the tree geometry of §2.3 and does not depend on the Fourier realization on the finite state space. For general β, the thermal occupation of the environment makes the aggregated up/down rates satisfy $\Gamma_\uparrow/\Gamma_\downarrow = p^{-\beta}$.

In the limit $K \to \infty$, $\beta = 1$, the Haar shell masses of §1.1 agree with the Gibbs shell distribution. The corresponding Haar constant vector expands over shells as $\sum_j \sqrt{\pi_j}\, e_j$ and satisfies $\langle V \rangle = p^{-1/2}$; the Gibbs state instead satisfies $\langle V \rangle = 0$. Equal shell occupations do not yet yield a thermal state: the shell-average coherences must still be damped by the dissipative dynamics.

3.3 The environment carrier

The environment mode of the collision model is still an abstract two-level degree of freedom. Its physical carrier must exchange energy $\log p$, preserve the character charges, and carry the phase records away from the local system; these three conditions jointly restrict the candidates.

First, the conservation screening. The raising isometry V commutes with all unit scalings and preserves every multiplicative character label; the pointwise form of the position kernel preserves the charges as well—both the ascent $x \mapsto px$ and the uniform fiber apportionment of the descent commute with M_u (§4.1): cross-shell jumps carry no character charge. An environment quantum that exchanges energy with these jumps while respecting the conservation laws must likewise carry no character charge: charged channels (exchange quanta with nontrivial character labels) are excluded by the coordinate-wise conservation laws.

Second, the energy screening. The energy-conserving exchange coupling requires the environment mode's frequency to equal $\log p$ exactly; a carrier with an intrinsic gap (mass) cannot match frequencies below its gap and cannot serve all prime directions simultaneously (the lowest frequency is $\log 2$; that a single carrier must serve all directions follows from the common-environment conclusion, §3.4); under the energy-scale identification of [30], the particle spectrum contains no massive candidate with gap below that value; a massless carrier faces no such restriction.

Third, the record screening. Relaxation to equilibrium requires the environment to carry the energy and phase records out of the system, never to return (the refreshing condition); only an escaping, unconfined carrier can hold permanent records.

The intersection of the threefold screening can be checked item by item against the boson identification of the Bost–Connes state space[30]: the massless ones are only two—the neutral quantum of the unbroken channel, and the color-channel quanta (gluons); the latter are confined, provide no storable record states, and are excluded by the record screening; fermionic candidates never enter the list—the repeated emission and absorption of a single quantum requires Bose statistics. The intersection of the three conditions is the massless neutral quantum of the unbroken channel, which under the particle identification of [30] is the photon; the leak probability α (the probability of releasing one quantum to the environment per dwell cycle) takes the value of the electromagnetic coupling. The conservation laws permit this coupling and exclude the remaining candidates.

$$\boxed{\text{Through the threefold screening, under the identification of the equilibrium framework, the thermalization carrier is uniquely the photon; the leak probability } \alpha \text{ is the electromagnetic coupling.}} \tag{50}$$

That the cross-shell coupling is nonzero—that an interaction operator with nonvanishing cross-shell matrix elements exists—is the physical input of the mechanism; the weak-coupling limit uses only $\alpha \ll 1$.

3.4 Joint relaxation of many prime directions

The collision model of a single prime direction cannot by itself explain why the whole system shares one temperature. Whether different prime directions can exchange energy directly, and whether a common environment can deliver each direction its own Boltzmann ratio, depends on whether their Bohr frequencies admit exact resonances.

With all prime directions present, the full Hamiltonian is

$$H_{\mathrm{tot}} = \sum_q (\log q)\, N_q, \tag{51}$$

with the N_q pairwise commuting; the KMS state factorizes along primes: in the convergence domain $\beta > 1$ of the Euler product its operator form is $\varphi_\beta = \bigotimes_q \varphi_{\beta,q}$ (at the pole $\beta = 1$ no global Gibbs density matrix exists, and the factorization persists only in the sense of quasi-local product states), and a single inverse temperature β simultaneously locks the Boltzmann ratio $q^{-\beta}$ of every direction. Fixing the prime p means tracing out the other factors; the complement of the p-local system then exists and is infinite-dimensional; records stored in the complement commute with all p-local evaluation operators and persist to the end of the process; the complement provides the algebraic characterization of record capacity and location, while the physical carrier of the records is the photon field (§3.3), and the exclusion of recurrence is borne by environment refreshing and the integrability of correlation functions (§3.1, §3.5). The environmental capacity demanded by the record screening of §3.3 is thereby supplied. The pole of the Euler-product partition function at $\beta = 1$ is produced jointly by all primes (§1.1).

There is no direct energy-exchange channel between prime directions. The Bohr frequency of an arbitrary cross-direction transfer has the form

$$\Delta E = \sum_q a_q \log q = \log \prod_q q^{a_q}, \qquad a_q \in \mathbb{Z}, \tag{52}$$

which vanishes if and only if $\prod_q q^{a_q} = 1$, and unique factorization forces all $a_q = 0$: no exact resonant process can raise one prime direction while lowering another.

$$\boxed{\text{No exact resonance channel exists between prime directions; joint relaxation can only proceed through a common environment.}} \tag{53}$$

By (53) together with the collision assumption that the environment modes of different directions are refreshed independently, the relaxations of the prime directions are mutually independent and couple only through the environment: each direction collides at its own frequency $\log q$ with the corresponding modes of one and the same neutral massless environment, and each converges to the bath temperature. The joint stationary state is therefore the tensor product of the local Gibbs factors at one β, and the joint relaxation spectrum is the union of the directional spectra: the overall gap is set by the slowest direction. The structure can be checked directly in a controlled realization: under alternating collisions with two environment modes at the same temperature (frequencies $\log 2$ and $\log 7$), the deviation of the joint stationary state from the product of the two local Gibbs factors is at the 10^{-15} level, the ratios $1/2$ and $1/7$ appear simultaneously, and the mutual information is at the 10^{-17} level (for script availability of the numerical checks see §6).

The no-resonance theorem excludes exact resonances; at finite coupling, non-secular virtual processes face near-resonant small denominators. The detuning of an (a, b)-order cross-direction process, $\delta = |a \log q_1 - b \log q_2|$, carries a Baker-type lower bound for linear forms in logarithms[27]: $\delta \geq c\, B^{-\mu}$ ($B = \max(a, b)$, with μ determined by the irrationality measure of $\log q_1 / \log q_2$); the number of order-B process paths is controlled by the bounded degree of the transition graph (of type $\leq (p+1)^B$), each order-B amplitude is at most $(CgB^{\mu}/c)^B$, and the optimal truncation of the perturbation series gives the upper bound $\exp(-c'\, g^{-1/\mu})$ for the cross-direction leak rate: the energy isolation between prime directions holds over exponentially long times. This argument is a Nekhoroshev-type outline[28]; the line-by-line bookkeeping of constants and exponents is not carried out in this paper. The irrationality measure serves here as the isolation exponent, and the protection of unique factorization extends to the non-secular layer through number-theoretic lower bounds.

The common environment therefore supplies one and the same inverse temperature β to all prime directions. The discrete collisions of the directions are already mutually compatible; what remains is whether they jointly admit a continuous-time limit.

3.5 The GKSL limit

The collision channel describes environment refreshing in discrete steps, while the relaxation spectrum requires continuous physical time. The continuum limit must simultaneously control the collision strength, the environment correlation time, and the system's Bohr frequencies; under these conditions the repeated-interaction model and the continuous bath give the same generator (up to the Lamb shift; for exchange coupling that term is diagonal and commutes with H, does not affect occupations or the rate ratio, and vanishes exactly in the repeated-interaction limit). The repeated-interaction limit[7] acts on (36); the Davies limit of a continuous bath[4] acts on the same Bohr-frequency decomposition. The convergence conditions of the latter are:

Condition 1: Unique system Bohr frequency. The coupling contains only the nearest-neighbor operators $V, V^{\dagger}$, whose transition frequencies are $\pm \log p$; the equal level spacing (§1.1) merges all nearest-neighbor transitions into a single frequency sector, so secular averaging splits off no extra sector; the exchange coupling has no diagonal component, so no zero-frequency sector appears;

Condition 2: Integrable environment correlations. In the collision picture, distinct incident modes are independent and the environmental correlation vanishes beyond adjacent collisions; in the continuous-bath picture one must assume that the environmental spectral density is smooth and nonvanishing at $\log p$ and that the correlation function is integrable—this spectral-class condition is a physical input, compatible with the carrier identification (§3.3). A single unrefreshed two-level mode fails this condition and therefore produces only coherent exchange, not thermalization;

Condition 3: Weak-coupling scaling. Taking $g \to 0$ with $g^2 t$ fixed, the Davies weak-coupling limit[4] yields convergence of the reduced dynamics to a GKSL semigroup, with the jump-rate ratio locked by the KMS condition to

$$\frac{\Gamma_\uparrow}{\Gamma_\downarrow} = e^{-\beta \log p} = p^{-\beta}. \tag{54}$$

The total rate $\kappa = \Gamma_\uparrow + \Gamma_\downarrow$ is not constrained by the equilibrium structure and serves as the overall rate unit of the dynamics. The resulting GKSL generator is the right-hand side of (46), denoted L; its restriction to the shell occupations is the continuous form of (41),

$$\dot{q}_j = \Gamma_\uparrow q_{j-1} + \Gamma_\downarrow q_{j+1} - (\Gamma_\uparrow + \Gamma_\downarrow) q_j \tag{55}$$

(reflecting endpoints), whose unique stationary distribution is (6). The decay of inter-shell coherence comes from the same set of exchange histories (44), so occupation transport and decoherence need not be added separately. This yields the continuous-time generator L; its spectra on the various coarse-grained readouts determine the actual relaxation hierarchy.

4 Relaxation dynamics

The approach to equilibrium is not a single exponential process. The shell occupations, shell-average coherences, character moments, and generation labels of an initial state are driven by one and the same collision dynamics, yet can survive to different time scales. The late-time behavior is governed by the slowest nonstationary readout among them; and when the environment temperature varies, the equilibrium state itself moves in time. All rates below are in units of κ.

4.1 The coarse-grained relaxation hierarchy

Energy distribution, channel information, and generation labels are different readouts of one quantum state. They are not complete coordinates of ρ_0; the shell occupations, the shell-average coherences, and the character moments of the unit shell each form closed equations, while the channel components of lower conductors form finite-dimensional tower equations (§4.2), so the relaxation times can be compared class by class. Given an initial state ρ_0, take the following four coarse-grained readouts. The shell occupations

$$q_j(0) = \mathrm{Tr}(P_j \rho_0) \tag{56}$$

record the distribution of energy along the valuation direction; the off-diagonal elements $C_{jk}(0) = \langle e_j | \rho_0 | e_k \rangle$ $(j \neq k)$ of the shell-average projection $\rho_0^{\mathrm{shell}} = P_{\mathrm{shell}} \rho_0 P_{\mathrm{shell}}$, with $P_{\mathrm{shell}} = \sum_j |e_j\rangle\langle e_j|$, record the coherences between shell-average states; the nontrivial multiplicative-character moments $c_\chi(0) = \sum_x \bar{\chi}(x) q_0(x)$ of the position-diagonal distribution $q_0(x) = \langle x | \rho_0 | x \rangle$ record the channel structure. For the nonzero-residue directions fix a coarse label map $g(x)$ and set

$$Q_g = \sum_{x \neq 0:\, g(x) = g} |x\rangle\langle x|, \qquad w_g(0) = \mathrm{Tr}(Q_g \rho_0), \tag{57}$$

the zero element carrying no label; w_g records the generation-label weights.

One and the same exchange process is read out in two representations. The shell-average representation takes $V, V^\dagger$ as jump operators (§3.1), acts on the shell-average subspace, and carries the shell occupations and the shell-average coherences; the position representation takes the pointwise form of the cross-shell exchange: ascent sends x to px (rate $\Gamma_\uparrow$), descent sends y uniformly to its p ascent preimages (rate $\Gamma_\downarrow$; descent from the zero element is shared equally among the $p-1$ nonzero preimages, consistent with the reflecting boundary of (55)). The position kernel satisfies pointwise detailed balance $\mu(x)\Gamma_\uparrow = \mu(px)\Gamma_\downarrow/p$, and its stationary distribution is intra-shell uniform times the shell weights of (6); summing over shells, the two representations aggregate into the same birth–death chain (55): the full shell occupation $\mathrm{Tr}(P_j\rho)$ of §1.1 is the q_j of (55) in the position representation, and the shell-average component $\langle e_j|\rho|e_j\rangle$ is its counterpart in the shell-average representation. The reduced equations of the character moments and of the generation labels are both computed in the position representation. The position kernel, the shell-average generator, and the label bottleneck give, respectively, the closed reduced equations of these readouts, and the evolution exhibits five stages in time.

Stage 1: collisions establish probability flux and output phase records. By (55), whenever $q_{j+1}/q_j \neq p^{-\beta}$ there is a net probability flux between adjacent shells. Meanwhile the absorption and emission records of (44) enter the environment, and the off-diagonal elements C_{jk} of the shell-average state contract under the reduced dynamics. Occupation transport and shell-average decoherence arise from the same quantum exchange process; the spectral gap controls the convergence of the shell occupations, and the rates of the off-diagonal elements are fixed by the same finite-dimensional GKSL generator.

Stage 2: the shell occupations approach the Gibbs ratio. At fixed β, symmetrizing the birth–death chain with respect to the stationary measure, its mode expansion is

$$q(t) = \pi_\beta + \sum_{a\geq 1} A_a e^{-\lambda_a t} u_a, \tag{58}$$

where π_β is (6), the u_a are nonstationary modes, and $\lambda_a > 0$. The slowest nonstationary mode is controlled by the spectral gap, and the relaxation time of the shell occupations is

$$\tau_{\text{shell}} \sim \frac{1}{\kappa\,\mathrm{gap}_\beta}. \tag{59}$$

For $t \gg \tau_{\text{shell}}$ the energy distribution is close to the Gibbs ratio and the details of the initial state along the valuation direction have been erased.

Stage 3: the channel structure of the position-diagonal distribution decays layer by layer in the conductor. Once the shell occupations are near equilibrium, the primitive character moments of the unit shell (conductor p^K) decay at a single rate; §4.2 computes fiber by fiber

$$c_\chi(t) = e^{-\nu_\beta t} c_\chi(0), \qquad \nu_\beta = \Gamma_\uparrow = \frac{\kappa}{1+p^\beta}, \tag{60}$$

so that at $\beta = 1$, $\tau_{\text{char}} = (p+1)/\kappa$. The channel components of lower conductors p^{K-j} $(j \geq 1)$ do not close into single modes; their closed objects are cross-shell towers of dimension $j+1$, whose slowest modes slow down by p^{-j} layer by layer (§4.2). At general temperature the time scales are set by the relative size of ν_β and gap_β. At $\beta = 1$ and $p \geq 5$, $\mathrm{gap}/\nu = (\sqrt{p}-1)^2 > 1$, so the shell occupations relax before the unit-shell channels; the finite-depth shell-occupation time is $\tau_{\text{shell}}^{(K)} = 1/(\kappa\,\mathrm{gap}_K)$.

Stage 4: the generation labels form the longest-lived slow variable. Cross-shell jumps preserve the coarse residue-field label, and only after reaching the deepest shell can transfer between labels occur. The mixing rate is ($\beta = 1$; for the Arrhenius form at general temperature see §4.3)

$$\mathrm{gap}_{\mathrm{sel}} = \Theta\big(p^{-(K-1)}\big)\kappa, \qquad \tau_{\mathrm{sel}} = \Theta\big(p^{K-1}\big)\kappa^{-1}. \tag{61}$$

At finite K, if the zero-element transfer kernel is irreducible among all labels, the labels still mix at very long times. This statement applies to the label-reduced chain; density-matrix components not covered by these readouts are outside the statement. The generation labels are the channel combinations of the lowest conductor p, with the deepest tower and the smallest slow mode: between adjacent rungs of the conductor ladder there are time-separation windows, and in absolute duration the widest window lies between the two lowest rungs. Writing τ_{sel} for the label mixing time, at deep truncation there is a time window

$$\tau_{\mathrm{char}} \ll t \ll \tau_{\mathrm{sel}}, \tag{62}$$

inside which, for $K = 2$, the shell occupations and all nontrivial channels have thermalized and only w_g remains nearly constant; for $K \geq 3$ the window still contains unrelaxed components of intermediate conductors, and the generation labels remain the slowest among them, behaving as effective superselection variables. Their mixing rate will be fixed by the deep-shell bottleneck in §4.3. The relaxation order of the system is therefore

$$\boxed{\begin{array}{c}\text{fast relaxation of shell occupations (shell-average coherences decay concurrently)}\\ \longrightarrow\ \text{channels relax layer by layer in the conductor}\\ \longrightarrow\ \text{extremely slow mixing of generation labels (lowest conductor)} \quad (\beta = 1,\ p \geq 5)\end{array}} \tag{63}$$

Stage 5: the fixed target generalizes to a moving target. At fixed environment temperature the shell-average semigroup satisfies

$$\frac{d}{dt} S(\rho_t^{\mathrm{shell}} \| \rho_\beta^{\mathrm{shell}}) \leq 0, \tag{64}$$

the relative entropy giving the direction from a shell-average initial state to the fixed Gibbs state. If the environment cools slowly, the target becomes $\rho^{\mathrm{shell}}_{\beta(t)}$. After an initial boundary layer $t \gtrsim (\kappa\, \mathrm{gap}_\beta)^{-1}$, the adiabatic estimate at finite K and on a bounded β interval is

$$\|\rho_t^{\mathrm{shell}} - \rho^{\mathrm{shell}}_{\beta(t)}\| \leq C e^{-\kappa \int_0^t \mathrm{gap}_{\beta(s)} ds} \|\rho_0^{\mathrm{shell}} - \rho^{\mathrm{shell}}_{\beta(0)}\| + \frac{C}{\kappa\, \mathrm{gap}_{\min}} \sup_{s \leq t} \big\|\dot\beta(s)\, \partial_\beta \rho^{\mathrm{shell}}_{\beta(s)}\big\|, \tag{65}$$

where the constant C depends on K and on the chosen temperature interval. The relative-entropy derivative with a moving target contains in addition the driving term $\partial_t \log \rho_{\beta(t)}$, so monotonicity at fixed temperature does not directly transfer to the cooling segment. The full trajectory splits into two layers: at short times the state approaches the equilibrium of the current temperature; at long times the equilibrium itself moves along the cooling curve. The relaxation spectrum of the generator at fixed temperature must be found first, before one can judge whether the internal dynamics tracks the latter scale fast enough.

4.2 The relaxation spectrum

The speed at which the shell occupations approach the Gibbs distribution is controlled by the slowest nonstationary mode, not by a single empirical damping constant. Determining this time scale requires the nontrivial spectrum of the shell-chain generator.

The restriction of L to the shell occupations can be diagonalized explicitly, with all relaxation rates in closed form. Insert the trial function $f_j = p^{js}$ into the normalized adjacency operator at $\beta = 1$: $[f_{j+1} + p\, f_{j-1}]/(p+1) = p^{js}\,(p^s + p^{1-s})/(p+1)$, i.e. eigenvalue $(p^s + p^{1-s})/(p+1)$; after symmetrization with respect to the stationary measure the operator is self-adjoint, the spectrum is a real interval, and its parametrization is exactly $s = \frac{1}{2} + i\vartheta$ (the standard form of spherical-function spectral theory on trees[21]): $p^s + p^{1-s} = 2\sqrt{p}\cos(\vartheta \log p)$ is real. $s = 1$ gives eigenvalue 1, the stationary direction; all nontrivial relaxation modes lie on $\operatorname{Re} s = 1/2$, with dispersion

$$\lambda(\vartheta) = \frac{2\sqrt{p}\,\cos(\vartheta \log p)}{p+1}, \qquad \vartheta \in \left[0, \tfrac{\pi}{\log p}\right], \tag{66}$$

a relaxation band $[\,1 - \frac{2\sqrt{p}}{p+1},\, 1 + \frac{2\sqrt{p}}{p+1}\,]\kappa$ of rates; the band parameter ϑ labels the oscillation of the eigenfunctions along the shell coordinate, with frequencies quantized in $\log p$. The spectral gap

$$\boxed{\text{gap} = 1 - \frac{2\sqrt{p}}{p+1} = \frac{(\sqrt{p}-1)^2}{p+1}} \tag{67}$$

has as its complement $2\sqrt{p}/(p+1)$ the Kesten spectral radius of the random walk on the $(p+1)$-regular tree[17]. The Alon–Boppana bound[18, 19] says that the second eigenvalue of any finite $(p+1)$-regular graph is at least $2\sqrt{p} - o(1)$: (67) is the largest spectral gap permitted by this family of geometries, and the relaxation mixes at the universal ceiling rate of regular-graph spectral theory (finite graphs attaining the bound are the Ramanujan graphs[20]). The gap of the finite-depth truncated chain lies above (67) and decreases monotonically to it with depth—in closed form $1 - \frac{2\sqrt{p}}{p+1}\cos(\pi/n)$ (n the chain length; the Alon–Boppana bound concerns the regular-graph family, while the truncated chain is non-regular at its two ends, and the boundary effect raises its gap above the limiting value); for the numerical sequence see §6.

The closed form of the gap at general temperature is $\text{gap}_\beta = 1 - 2p^{\beta/2}/(1 + p^\beta)$; as β grows, the up/down rate ratio narrows as $p^{-\beta}$ and the gap widens.

Beyond the shell occupations, the relaxation of the intra-shell degrees of freedom is computed case by case in the position representation. First take as observable a primitive Dirichlet character χ mod p^K on the unit shell S_0 (viewed as a function on the full ring, zero outside), and write its component $c_\chi(t) = \sum_x \bar{\chi}(x)\, q_t(x)$. Outflow: the ascent step moves support from S_0 to S_1, where χ vanishes, so c_χ drains at the ascent rate; states in S_0 contain no factor p, have no quantum to emit, and possess no descending channel. Inflow: when the distribution on the shell S_1 above descends into S_0 it is spread uniformly over the p ascent preimages, and a primitive character sums to zero along every fiber: the inflow is annihilated pointwise. The component equation therefore closes as $\dot{c}_\chi = -\Gamma_\uparrow c_\chi$, with linear damping

$$\nu_\beta = \Gamma_\uparrow = \frac{\kappa}{1 + p^\beta}, \qquad \nu = \frac{\kappa}{p+1} \quad (\beta = 1), \tag{68}$$

degenerate across all channels of conductor p^K: the computation used no specific values of χ, only support on the unit shell and zero fiber sums. The ratio of damping to gap is

$$\frac{\nu}{\text{gap}} = \frac{1}{(\sqrt{p}-1)^2}\,. \tag{69}$$

The unit-shell channels lie below the relaxation band: character degrees of freedom outlive the shell occupations.

Channels of lower conductor do not close into single modes. Fixing a multiplicative character of conductor p^{K-j} ($j \geq 1$), its lifts appear on shells 0 through j, and the position representation couples these $j+1$ components into a tridiagonal tower: adjacent shells inside the tower are connected both ways by $(\Gamma_\uparrow, \Gamma_\downarrow)$, the top (shell j) leaks outward with $\Gamma_\uparrow$, and descent from above is still annihilated by the fiber sums. The position kernel is therefore completely block-diagonalized by the conductor of the multiplicative character: the tower of conductor p^m has dimension $K-m+1$ ($1 \leq m \leq K$), and the tower of the trivial character is the shell-occupation chain itself. The decay rates of the two-dimensional tower ($j=1$) are in closed form:

$$\lambda_\pm = \frac{(p+2) \pm \sqrt{p^2+4p}}{2(p+1)}\,\kappa \qquad (\beta = 1), \tag{70}$$

the slow mode λ_- being $(9-\sqrt{77})\kappa/16 = 0.014065\,\kappa$ at $p=7$; the slowest mode of a deep tower is asymptotically $\Gamma_\uparrow(1-1/p)^2 p^{-j}$. The channel structure thus relaxes in conductor layers: each drop of one conductor step slows the slowest mode by a factor of about p.

The conductor block structure of the position kernel closes into one characteristic-polynomial identity. Writing T for the position kernel, B_d for the $(d+1)$-dimensional tower block of conductor p^{K-d}, B_K^{pr} for the principal block containing the zero element (the shell-occupation chain), and μ_m for the number of multiplicative characters of conductor exactly p^m,

$$\det\bigl(zI - T\bigr) = \chi_{B_K^{\text{pr}}}(z) \prod_{a=0}^{K-1} \chi_{B_a}(z)^{\mu_{K-a}}, \tag{71}$$

and the tower block B_d depends only on the tower depth $d = K-m$, not on K itself. This yields the cross-depth nesting of the spectrum: all non-principal eigenvalues of the depth-$(K-1)$ system reappear unchanged in the depth-K system (multiplicities of the conductor-p^a tiers with $a \geq 2$ are magnified exactly by p; the shallowest tier grows from $p-2$ to $(p-1)^2$), while the new tower contributes one slower out-of-band mode and several new in-band lattice points. The principal-block (shell-chain) spectrum moves with depth; its nonzero spectrum is numerically exactly the cosine lattice $1 - \frac{2\sqrt{p}}{p+1}\cos(k\pi/n)$ ($k = 1, \ldots, n-1$, n the chain length, verified to 10^{-15}), so the gap has the closed form $1 - \frac{2\sqrt{p}}{p+1}\cos(\pi/n)$ and converges to (67) monotonically from above with the chain length: the chain-length sequence of §6.3 is thereby secured in closed form. The in-band eigenvalues are asymptotically the cosine lattices of the towers, and their geometrically weighted mixture over conductor multiplicities remains atomic as $K \to \infty$: the relaxation response is a countable comb of relaxation lines.

The relaxation spectrum carries one further layer of rigidity against microscopic details of the realization. In the definition of the position kernel, the apportionment of the descent among the p ascent preimages is a microscopic degree of freedom; the following statement shows this freedom to be pure gauge:

Under the three constraints of tree structure, fixed escape-rate profile, and no holding at the zero element, the characteristic polynomial is independent of all branch weights.	(72)

The weights may take arbitrary pointwise values (even complex ones, with the weights on each descent fiber summing to one), and the rates may depend on the shell index; the conclusion holds in all these cases. If the rates vary with position inside a shell, rigidity fails. The outline of the proof has three steps. Step 1: for a ball $B = x + p^i\mathbb{Z}/p^K\mathbb{Z}$ define its shadow $s(B)$ as the point to which the whole ball collapses under repeated compression by the ascent flow; let $\mathfrak{A}_a$ be the span of indicator functions of balls whose shadow valuation is at least $a+1$, and F_a its annihilator. Then the flag $0 = F_{-1} \subset F_0 \subset \cdots \subset F_K$ is preserved by all weight kernels jointly: on indicators of balls whose radius is at least that of a descent fiber, the kernel action is weight-independent; the single-point balls are subsumed by a conditioning step—elements of F_a vanish at the deeper point pz, and the weight-sensitive port reads zero exactly there. Step 2: each column of the difference of two weight kernels is a zero-sum vector along one descent fiber, and the difference operator maps F_a into F_{a-1}: strictly nilpotent. Step 3: all weight kernels induce one and the same map on the graded pieces of the flag, so the characteristic polynomials agree block by block, and (71) holds for arbitrary weights. Physical reading: imperfections of the branch weights can only move amplitude one way, from coarse conductor sectors into finer ones, never back, and therefore cannot move any eigenvalue. Rigidity protects the spectrum; the intra-shell uniformity of the stationary state is destroyed by non-uniform weights, so preparation and readout protocols premised on intra-shell uniformity are outside the protection. That the zero-element transfer involves no holding in place is precisely the position-kernel convention of §4.1; numerically this convention is necessary for rigidity—allowing a self-loop at the zero element moves the spectrum.

4.3 Effective superselection

Cross-shell jumps preserve the residue-field label, but at the zero element that label ceases to exist. The generation labels are therefore neither strictly conserved nor mixed at generic shell rates: changing a label requires first reaching the deepest valuation and then passing through the zero element, so the lifetime is controlled by this probabilistic bottleneck. The two readings of the bottleneck agree. First, the cut (conductance) estimate: the stationary probability flux through the cut is the deepest-shell weight times the ascent rate ($\beta = 1$), $\pi_{K-1}\Gamma_\uparrow \sim p^{-(K-1)}\kappa/(p+1)$, and the label-block indicator as trial function gives an upper bound of the same order. Second, the explicit spectrum: the channel combination of the generation labels belongs to the lowest conductor p, whose tower is a K-dimensional tridiagonal matrix; the explicit approximation to its slowest eigenvalue is the Newton step $\lambda = \Gamma_\uparrow/w_{K-1}$ (the first Newton iterate from the origin, i.e. $P(0)/P'(0)$ of the characteristic polynomial, with relative error $O(p^{-\beta(K-1)})$, exact in the deep-tower limit), where $w_j = 1 + (1+p^\beta)w_{j-1} - p^\beta w_{j-2}$ ($w_{-1} = 0$, $w_0 = 1$) has the closed solution $w_d = 1 - R^2/(R-1)^2 + R^{d+2}/(R-1)^2 + d/(1-R)$, $R = p^\beta$. The exact value at $K = 2$, $\beta = 1$ is the λ_- of (70); exact diagonalization gives, for $p = 7$ and $K = 2, 3, 4$, the values 0.014065, 0.001901, 0.000268, with successive ratios approaching $1/p$. The spectral identity of the slow mode is the remnant of charge conservation: the boundary state of the semi-infinite tower, in per-site intra-shell occupation, is $f_j = (p^{1-\beta})^j$ (channel-moment components $p^{-\beta j}$) and carries an exact zero eigenvalue—the character charge is conserved at infinite depth; finite depth splits this zero mode into an exponentially small leak rate, asymptotically of Arrhenius form

$$\boxed{\mathrm{gap}_{\mathrm{sel}} = \Gamma_\uparrow\big(1 - p^{-\beta}\big)^2 e^{-\beta E_b}\big(1 + o(1)\big), \qquad E_b = (K-1)\log p,} \tag{73}$$

The activation energy is the conductor depth times the energy quantum: changing a label requires sending $K-1$ quanta against the Boltzmann weights toward the zero element; the K-th quantum (the escape through the zero element) hides in the prefactor $\Gamma_\uparrow$, so the measured slope in units of κ is $K\log p$, while $(K-1)\log p$ is the activation energy relative to the unit-shell damping $\Gamma_\uparrow$. At $\beta=1$ this degenerates to the critical reading $\mathrm{gap}_{\mathrm{sel}} = (1-1/p)^2 p^{-(K-1)}\kappa/(p+1)\cdot(1+o(1))$; temperature enters the exponent, and the generation structure becomes exponentially more stable at low temperature. Inside the time window $\tau_{\mathrm{char}} \ll t \ll \tau_{\mathrm{sel}}$ the generation labels form effective superselection variables; their leakage is suppressed by the deep-shell bottleneck. Finite K corresponds to effective superselection of finite lifetime: the labels can still mix at extremely long times. The shape of the slow-mode wavefunction switches with temperature: for $\beta > 1$ it localizes at the unit-shell end (label memory carried by shallow shells), for $\beta < 1$ at the zero-element end, and at $\beta = 1$ it is exactly uniform. The critical temperature acquires here one more identity: the pole of the partition function, the Haar stationary measure, and the localization–delocalization transition of the slow mode coincide.

The rates of all readouts assemble into a conductor-layered ladder ($\beta = 1$):

$$\underbrace{\Theta(p^{-(K-1)})\,\kappa}_{\text{conductor } p \text{ (generation labels)}} \quad \ll \;\cdots\; \ll \underbrace{\tfrac{(1-1/p)^2}{p+1}\,p^{-1}\kappa}_{\text{conductor } p^{K-1}} < \quad \frac{\kappa}{\underbrace{p+1}_{\text{conductor } p^K}} \quad < \underbrace{\Big[1-\tfrac{2\sqrt{p}}{p+1},\,1+\tfrac{2\sqrt{p}}{p+1}\Big]\kappa}_{\text{shell-occupation band}}, \tag{74}$$

with adjacent rungs in ratio about $1/p$ (the conductor-p^{K-1} rung uses the deep-tower asymptotics; its exact value is the λ_- of (70)); the last inequality holds for $p \geq 5$, while at $p = 3$ the damping of the conductor-p^K channels lies above the lower band edge. Time reading of the ladder: the arithmetic resolution still alive at time t is $m^*(t) = K - \log_p(\kappa t) + O(1)$ ($\beta = 1$; at general temperature the slope becomes $\beta^{-1}\log_p$; the constant term in the deep-tower limit is $-\log_p[(1-1/p)^2/(p+1)]$). The thermalization clock erases conductor depth at a logarithmic rate, losing one rung for every factor of p in time. The rate ladder describes how fast equilibrium is approached, but does not select a direction of time; the latter is fixed by the monotonicity of the relative entropy.

4.4 Relative entropy and the arrow of time

The spectral gap says that off-equilibrium modes decay, but by itself cannot distinguish the two directions of time. For the detailed-balance semigroup on the shell-average sector, the relative entropy supplies a monotone state function (the classical chain in the position representation is stationary at the full Gibbs distribution, and the same argument runs in parallel):

$$\frac{d}{dt}\,S\big(\rho_t\,\|\,\rho_\beta^{\mathrm{shell}}\big) \leq 0. \tag{75}$$

Two facts carry the proof: the shell Gibbs state $\rho_\beta^{\mathrm{shell}}$ is stationary for the semigroup T_t ((54) and (55)); and relative entropy does not increase under any completely positive trace-preserving map (the data-processing inequality[8, 9]). Hence $S(T_t\rho\,\|\,\rho_\beta^{\mathrm{shell}}) = S(T_t\rho\,\|\,T_t\rho_\beta^{\mathrm{shell}}) \leq S(\rho\,\|\,\rho_\beta^{\mathrm{shell}})$; in the case of equality, the recoverability underlying the data-processing inequality (Petz) places ρ inside the decoherence-free subalgebra, which is trivial by primitivity (irreducible jump operators and faithful stationary state, §3.1), so equality holds only at $\rho = \rho_\beta^{\mathrm{shell}}$. The approach to equilibrium is therefore monotone in the strict Lyapunov sense, the strictness coming from primitivity together with detailed balance: the direction of time is given by the sign of the entropy production. This yields nonnegative entropy production and the corresponding direction of time; for the systematic theory of the nonnegativity of entropy production in GKSL dynamics see Spohn[10].

The monotonicity has a finer, trajectory-level form. By (54), each jump and its reverse have rate ratio $e^{\beta\omega_k}$ (ω_k the net energy released to the environment at step k, absorption counted negative); multiplying along a trajectory (the waiting-time factors of the forward and reversed trajectories cancel), the conditional path probabilities with given endpoints satisfy $P[\omega \mid x_0]/P[\bar{\omega} \mid x_\tau] = e^{\beta Q(\omega)}$, with $Q(\omega)$ the total energy released to the environment along the trajectory: reversed trajectories are exponentially suppressed one by one. Taking the uniform reference initial distribution for the environmental entropy flow $\sigma = \beta Q(\omega)$, its distribution obeys the detailed fluctuation relation (for the family of related results see Crooks[14], Gallavotti–Cohen[15], and Jarzynski[16])

$$\frac{P(\sigma)}{P(-\sigma)} = e^{\sigma}, \tag{76}$$

exact class by class. Under the equilibrium ensemble the distribution of σ is strictly symmetric—no arrow at equilibrium, consistent with the equality case of (75): the direction lives only in relaxation away from equilibrium, expressed at the two levels of mean entropy production and conditional path ratios. In this model the rate ratio and the energy quantum are given by (54) and $\log p$, respectively.

The monotonicity and the fluctuation relation combine into a single corollary, stated at three levels. At the mean level, nonnegative entropy production (equation (75)) fixes the direction of relaxation; at the fluctuation level, the conditional path ratio and (76) refine the direction to individual trajectories; at the record level, the record of each step of entropy production is carried away by the post-collision environmental photon and never returns (environment refreshing, §3.1), so the physical bearer of irreversibility is identical with the thermalization carrier (§3.3). The first two levels issue from detailed balance; the third is the record form of the refreshing boundary condition; beyond that boundary condition no new input is introduced. The structural origin of the direction belongs to the equilibrium framework[30]; what this paper establishes is the dynamical theorem within the given structure:

$$\boxed{\text{The arrow of time is a theorem of this mechanism, not an assumption: mean entropy production, trajectory-wise fluctuations, and escaping records all point one way.}} \tag{77}$$

The time asymmetry of this theorem has a single entry point. The joint evolution is unitary, and the absorption and emission branches of a single collision are time reversals of each other (§3.1); H_{int} has real matrix elements in the product basis, the antilinear time-reversal operator Θ satisfies $\Theta U_\tau \Theta^{-1} = U_\tau^{-1}$, and the incident thermal state is Θ-invariant as well ($\eta_{\beta,p}$ diagonal and real). The asymmetry comes entirely from the refreshing condition, which has two clauses: the incident mode is uncorrelated with the system before the collision; the outgoing mode departs with its record and takes no further part in the evolution. If the departed modes were sent back coherently, together with all their entanglement with the system, and the inverse unitaries of the collisions were applied in reverse order, the process would retrace itself step by step: irreversibility is borne by this boundary condition, while the dynamics itself is reversible. The physical content of the condition is the record screening (§3.3)—escaping photons hold the permanent records; the exclusion of recurrence is borne by the environment's infinite capacity and the integrability of correlation functions (§3.4, §3.5); in the Davies continuous-bath picture the same condition appears as the global product initial state and the forward weak-coupling limit. The monotonicity holds for arbitrary initial states; nonzero entropy production requires the initial state to be off equilibrium, and during cooling the sustained entropy production is maintained by the leak drive (§4.5). The dividing line of (77) is therefore exact: the derivations inside the mechanism are theorems, and environment refreshing (the two clauses above) is the only time-asymmetric assumption in the paper.

A fixed prime direction still has a normalizable shell Gibbs distribution and a positive spectral gap at $K \to \infty$, $\beta = 1$. The divergence of the global Euler product at $\beta = 1$ belongs to the many-prime structure; and if the temperature itself varies in time, the stationary state moves with it.

4.5 Quasistatic cooling

The relative-entropy monotonicity above presupposes a fixed environment temperature. During cooling, the Gibbs state itself moves with $\beta(t)$; only when the internal relaxation is faster than the temperature drift does the system track this instantaneous equilibrium. The escaping, non-returning environment quanta (the photons) constitute an effective zero-temperature environment, and cooling is physically spontaneous emission into the vacuum; the system re-equilibrates at rate κ and leaks outward at rate $\gamma = \alpha\kappa \ll \kappa$, with α the leak probability of §3.3. $\beta(t)$ is fixed by the energy-flux balance, and the tracking error is estimated once the cooling equation is in place.

The starting point of cooling is selected by the energy-capacity structure. Once thermalization is established, the expected energy of the full system at the self-consistent temperature β is exactly the von Mangoldt series

$$E(\beta) = \sum_q \frac{(\log q)\, q^{-\beta}}{1 - q^{-\beta}} = \sum_{n \geq 2} \Lambda(n)\, n^{-\beta} = -\frac{\zeta'}{\zeta}(\beta), \tag{78}$$

strictly decreasing for $\beta > 1$ and divergent as $\beta \to 1^+$ through the pole of ζ: the self-consistency equation $E(\beta) = E_0$ has a unique solution for arbitrarily large initial energy E_0, and $\beta \to 1^+$ as $E_0 \to \infty$. The dependence of the starting point on the initial energy is therefore only logarithmic: any large E_0 reads out $\beta_0 \approx 1$, and the starting point is a reading of the energy-capacity structure; the energy of a single prime direction stays finite as $\beta \to 1$, the divergence being a collective effect of all primes.

First fix the generator in the occupation-number representation. Unique factorization

$$n = \prod_q q^{a_q}, \qquad a_q = v_q(n) \tag{79}$$

gives the state-space decomposition

$$\ell^2(\mathbb{N}^*) \cong \bigotimes_q \ell^2(\mathbb{Z}_{\geq 0}). \tag{80}$$

The full Hamiltonian (51) reads $H_{\text{tot}} = \sum_q (\log q) N_q$, and the $H = (\log p)N$ of §1.1 is its p-component. The shell shift V has unit nonzero matrix elements, corresponding to **constant-rate** birth–death with jump rates independent of occupation. If each excitation quantum couples to the environment separately, the jump matrix elements are $\sqrt{j}$, giving the **bosonic-rate** model. The latter requires per-quantum excitation labels and is treated as a different microscopic coupling; the two models share the same late-time cooling law.

Evaluating the zero-temperature outflow on the instantaneous Gibbs state gives the energy flux. The Gibbs occupation of direction q is geometric ($r_q = q^{-\beta}$), and the elementary identities $P(N_q \geq 1) = r_q$ and $\langle N_q \rangle = r_q/(1 - r_q)$ give the exact relation between the two models' energy fluxes ($E_q = (\log q)\langle N_q \rangle$, $\gamma_q = \alpha\kappa_q$):

$$\frac{dE_q}{dt}\Big|_{\text{const}} = -\gamma_q(\log q)\, r_q = \frac{dE_q}{dt}\Big|_{\text{bosonic}} \times (1 - q^{-\beta}), \qquad \frac{dE_q}{dt}\Big|_{\text{bosonic}} = -\gamma_q E_q. \tag{81}$$

The two models differ by the closure factor $(1 - q^{-\beta})$, which tends to one exponentially as β grows.

In the late low-temperature stage, the energy is dominated by the smallest prime direction: $E \simeq (\log 2)\, 2^{-\beta}/(1 - 2^{-\beta})$, and this segment involves the single direction $q = 2$. Quasistatic reparametrization gives the closed equations of the two models,

$$\frac{d\beta}{dt} = \frac{\gamma_2}{\log 2}(1 - 2^{-\beta})^2 \text{ (constant)}, \qquad \frac{d\beta}{dt} = \frac{\gamma_2}{\log 2}(1 - 2^{-\beta}) \text{ (bosonic)},$$

which agree asymptotically:

$$\boxed{\frac{d\beta}{dt} \to \frac{\gamma_2}{\log 2}, \qquad \beta(t) \simeq \beta_0 + \frac{\gamma_2}{\log 2}\, t, \qquad T(t) \propto \frac{1}{t} \ \ (t \to \infty).} \tag{82}$$

The late cooling law (β linear, $T \propto 1/t$) is therefore model-independent; the slope is fixed by the down-jump rate of the first excited state, which for both couplings equals $\gamma_2 = \alpha\kappa_2$ (κ_q being the rate unit of that direction; a general coupling $f(j)$ gives $\gamma_2 f(1)/\log 2$), and the rate is fixed by the $q = 2$ direction alone, independent of the others. The subleading correction factor discriminates: $(1 - 2^{-\beta})^2$ versus $(1 - 2^{-\beta})$, differing by about 3% at $\beta = 5$; the experimental form of this criterion is given in §6.2.

The critical neighborhood does not change this picture. As $\beta \to 1^+$, the energy reservoir is supplied jointly by all prime directions: truncating at $q \leq Q$, $E_0 = \sum_{q \leq Q} \log q/(q - 1) = \log Q + O(1)$ (Mertens[26]). By (81), the ratio of the two models' energy fluxes is the closure factor $(1 - q^{-\beta})$, which for $\beta \geq 1$, $q \geq 2$ lies between $1/2$ and 1: the fluxes are of the same order, $dE/dt = -\Theta(\gamma)E$, and the time to leave the critical neighborhood is $\Theta(\gamma^{-1} \log\log Q)$ for both models (assuming the directional rate units κ_q have uniform upper and lower bounds; the Θ constants depend on their ratio). The microscopic coupling type produces no scaling distinction in the critical neighborhood; all its discriminating signal sits in the mid-temperature correction factor (§6.2).

Finally, control the quasistatic error. With the two time scales $\kappa \gg \gamma = \alpha\kappa$, after the initial fast-relaxation layer the deviation of the shell distribution from the instantaneous Gibbs state is controlled by (65); at fixed finite K and on a bounded temperature interval, with $\dot{\beta} = O(\alpha\kappa)$ this error is $O(\alpha/\text{gap}_{\min})$ and the corresponding subleading correction to the energy flux is $O(\alpha^2)$. The estimate extends over the whole cooling course on two grounds: gap_β increases monotonically in β (so $\text{gap}_{\min} = \text{gap}_{\beta_0} > 0$ holds uniformly), and the norm of $\partial_\beta \rho_\beta^{\text{shell}}$ tends to zero as $\beta \to \infty$ (so the constant can be taken uniform). Under quasistatic cooling any time-evolution equation can be reparametrized by β, $\partial_t = \dot{\beta}\, \partial_\beta$, all coefficients rescaling by $\dot{\beta}$. The early stage and the late linear cooling are joined smoothly by the full shape of $E(\beta)$; the temperature does not reach zero in finite time. Equation (82) and the adiabatic error bound jointly determine the quasistatic cooling trajectory.

4.6 The mean-field limit of the resonance amplitudes

The GKSL equation acts on density matrices, while the equilibrium framework works with resonance amplitudes A_k. Passing between the two descriptions requires a closure of higher moments; discarding the connected correlations yields the mean-field equations, whose window of validity is set by the growth of those correlations.

The resonance amplitude $A_k = \langle a_k \rangle$ is the coherent amplitude of an intra-shell character direction; the mode operators a_k belong to the channel expansion of the equilibrium framework[30]. The lowest-order nonlinear (beyond quadratic) invariant permitted by the channel conservation laws (coordinate-wise conservation of character labels) is cubic, with structure constants the character convolution $f_{ijk} = \delta_{\chi_i \chi_j, \chi_k}$; the vertex carries the square root of the coupling, and the resonance condition is enforced by character multiplication. The cubic vertex belongs to the resonance-coupling layer of the equilibrium framework and is not generated by L.

Taking Heisenberg expectation values of the joint action of the open dynamics L and this coupling layer gives

$$\frac{dA_k}{dt} = -(\nu_k + i\Delta_k)A_k - ig\sqrt{\alpha_k}\sum_{ij} f_{ijk}\, \langle a_i a_j \rangle + J_k, \tag{83}$$

where Δ_k is the detuning of channel k, α_k its coupling constant (the leak probability α of §3.3 is its electromagnetic-channel value), and J_k a source term. The linear damping is taken to be the unit-shell channel value $\nu_k = \nu$ of (68): this identification is exact at the level of position-diagonal moments; applying it to coherent amplitudes is a convention for interfacing with the phenomenological equations of the equilibrium framework, and all conclusions of this subsection are stated under that convention.

The second moments can be written $\langle a_i a_j \rangle = A_i A_j + C_{ij}$. The mean-field truncation discards the connected part C_{ij} and, reparametrizing along the cooling trajectory $\partial_t = \dot{\beta}\, \partial_\beta$, gives

$$\dot{\beta}\, \partial_\beta A_k = -(\nu_k + i\Delta_k)A_k - ig\sqrt{\alpha_k}\sum_{ij} f_{ijk} A_i A_j + J_k. \tag{84}$$

If the equation is normalized as $\partial_\beta A_k + \widetilde{\nu}_k A_k + \sqrt{\widetilde{\alpha}_k}\sum f_{ijk} A_i A_j = \widetilde{J}_k$, then all tilde coefficients are the corresponding time coefficients divided by $\dot{\beta}$; they must not be mixed with the coefficients of the original time equation. Here f_{ijk} is the unique cubic invariant of the conservation laws, $g\sqrt{\alpha_k}$ the amplitude-level vertex, and ν_k the unit-shell channel damping (68).

The window of validity of the truncation is controlled by the growth of the connected correlations. Inside the window where damping and detuning are negligible the error depends only on the combination gt; the deviation between the three-mode full quantum dynamics and the mean-field trajectory obeys, to leading order, the self-similarity law $\mathrm{err}(g/2, 2t) = \mathrm{err}(g, t)$ (numerically 4.29% $\leftrightarrow$ 4.30%, 18.76% $\leftrightarrow$ 18.91%; the check is restricted to the three-mode system), and numerical fits support a bound of the form

$$\big|\langle a \rangle - A_{\mathrm{MF}}\big| \leq C\,(gt)^2 e^{C_1 gt} \tag{85}$$

(not proven in this paper), with effective window $gt \lesssim 0.5$ (error $< 5\%$). Outside the window the deviation is taken over by the connected correlations, which belong to the fluctuation-correction layer; the condensation readout uses occupations as order parameter, its definition does not rely on the amplitude layer, and the discarded connected part enters only its subleading corrections.

Two testable consequences follow (both under the interfacing convention above): near the vacuum ($A \to 0$) the damping returns to the channel-independent (68), and any channel-dependent linear damping conflicts with the mechanism; channel dependence of the damping can enter only through mode-coupling renormalization by the occupation distribution, of the form $\delta\nu_k \propto \sum_j |g\sqrt{\alpha} f|^2 \, n_j/\nu$.

The mean-field equations describe the coherent amplitudes of intra-shell characters; the joint expansion of mass generation and cross-channel transitions with the open dynamics is outside the scope of this paper. The uniqueness of the dilation data is treated in §5, the experimental tests in §6.

5 Uniqueness of the dilation structure

Existence does not exclude different dilations producing different system-side dynamics. Uniqueness splits into two layers here: the finite Heisenberg representation controls the realization of the Weyl automorphism, and the minimal unitary dilation controls the realization of the adjacent-shell shift in the larger space. The Hecke double coset only classifies the adjacent directions on the tree and plays a part in neither of the two uniqueness statements.

5.1 Uniqueness of the Weyl realization

If two unitaries realize the same Weyl automorphism, their difference must sit in the commutant of the finite Heisenberg representation. That commutant decides whether the Fourier realization introduces new physical degrees of freedom.

First fix the Weyl automorphism of (29):

$$F\varepsilon_r F^{-1} = T_{-r}, \qquad F T_c F^{-1} = \varepsilon_c. \tag{86}$$

Position phases and translations satisfy the Weyl commutation relations and generate the finite Heisenberg group, whose center acts on $\mathcal{H}_{p,K}$ through a fixed nontrivial character.

This Heisenberg representation is irreducible: an operator commuting with all ε_r is position-diagonal, and commuting also with all T_c makes it constant.

Suppose F' realizes the same automorphism. Then $F'\varepsilon_r F'^{-1} = F\varepsilon_r F^{-1}$ and $F' T_c F'^{-1} = F T_c F^{-1}$ show that the quotient $F'F^{-1}$ commutes with all ε_r, T_c; by irreducibility and Schur's lemma the quotient is a scalar (for the uniqueness of the finite Heisenberg representation see [23]). Any realization satisfying the above conjugation relations is

$$F' = e^{i\theta} F. \tag{87}$$

Imposing $F'^2 = \iota$ gives $e^{2i\theta} = 1$, compressing the continuous phase to a sign, $F' = \pm F$. The orientation convention of the normalized quadratic Gauss sum

$$\operatorname{Tr} F = p^{-K/2} \sum_x e^{2\pi i x^2/p^K} \tag{88}$$

(real and imaginary parts of $\operatorname{Tr} F$ both nonnegative; the value always lies in $\{1, i\}$) selects the sign of (28). This choice is a normalization convention of the Fourier kernel; all dynamical outputs depend only on moduli of matrix elements or the real/imaginary type of eigenvalues, and are invariant under $F \mapsto -F$.

5.2 Uniqueness of the minimal unitary dilation

One and the same shift operator can be embedded into different Hilbert spaces. The form of the auxiliary space may vary; the system embedding and its compressed matrix elements must not. Let U_1 and U_2 be two minimal unitary dilations of V, with embeddings I_1 and I_2. The unitary-equivalence property of minimal dilations[29] supplies a unitary

$$W : \mathcal{K}_1 \longrightarrow \mathcal{K}_2, \qquad WI_1 = I_2, \qquad WU_1 = U_2W. \tag{89}$$

Hence the two minimal dilations are unitarily equivalent once the system embedding is fixed, and the system-side compression (24) is the same.

In summary: the adjacent-shell shift V is fixed by (20); its minimal unitary dilation is unique up to unitary equivalence preserving the system embedding; and the Fourier element realizing (29) with $F^2 = \iota$ is unique up to sign, the sign being fixed by the Gauss-sum orientation convention of the Fourier kernel. The dynamics still contains the overall rate unit κ.

6 Experimental tests

Structural existence and uniqueness do not prove that these operators are realized in a physical system. Experiment must separate three physical claims: whether the finite Weyl conjugation has a physical realization, which microscopic type the dissipative coupling belongs to, and whether the relaxation spectrum can hit the closed-form rate ratios and time scales simultaneously.

6.1 The Majorana phase criterion

The overall sign of the Fourier element is a normalization convention, but whether its eigenvalue on the real quadratic character is real or imaginary does not change with that sign. If this character corresponds to a self-conjugate particle direction, its Fourier phase becomes a measurable particle property.

From $F^2 = \iota$ and $F^4 = 1$, the Fourier eigenvalues lie in $\{1, i, -1, -i\}$. The valuation lifts of the real quadratic character (Legendre symbol) χ of the residue field give explicit eigenvectors. Write $\hat{\chi}$ for the lift of χ to the unit shell ($\hat{\chi}(x) = \chi(x \bmod p)$, zero off the shell) and $\hat{w}$ for its mirror on the deepest nonzero shell ($\hat{w}(p^{K-1}u) = \chi(u)$, zero elsewhere). Summing in layers over $x \bmod p$, the inner geometric sums compress the Fourier image to the two ends: $F\hat{\chi} = p^{K/2-1}\tau(\chi)\,\hat{w}$, $F\hat{w} = p^{-K/2}\tau(\chi)\,\hat{\chi}$. The equal-weight combination of the two ends is therefore an eigenvector:

$$F\,v_\chi = \frac{\tau(\chi)}{\sqrt{p}}\,v_\chi, \qquad v_\chi = p^{(1-K)/2}\hat{\chi} + \hat{w}, \qquad \frac{\tau(\chi)}{\sqrt{p}} = \begin{cases} 1, & p \equiv 1 \pmod 4, \\ i, & p \equiv 3 \pmod 4, \end{cases} \tag{90}$$

where $\tau(\chi)^2 = \chi(-1)p$; at $K = 1$ one has $\hat{w} = \hat{\chi}$ and v_χ degenerates to a multiple of $\hat{\chi}$, while the opposite-sign combination $-p^{(1-K)/2}\hat{\chi} + \hat{w}$ (for $K \geq 2$) carries eigenvalue $-\tau(\chi)/\sqrt{p}$, of the same real/imaginary type. The type is decided by $\tau(\chi)^2 = \chi(-1)p$, i.e. by $p \bmod 4$; the sign of the Gauss sum[25] then selects which of the $\pm$ branches belongs to v_χ.

Under the standard-model identification of the Bost–Connes state space[30] (three generations correspond to the involution orbits of the unit group, gauge factors to the CRT factors of the residue field), the neutrino corresponds to the unique self-conjugate elementary direction, the real quadratic character; the way particle and antiparticle coincide is given by the self-conjugation structure of that direction. For $p \equiv 3 \pmod 4$ the Fourier eigenvalue of the self-conjugate direction is purely imaginary ($\pm i$): particle and antiparticle are identified through a quarter phase—the Majorana type; for $p \equiv 1 \pmod 4$ the eigenvalue is real and there is no such phase. The identification takes $p = 7 \equiv 3 \pmod 4$: under it, "the neutrino is a Majorana fermion" and "the eigenvalue of F on this direction is purely imaginary" correspond to one and the same reading ($\chi(-1) = -1$).

Neutrinoless double beta decay tests this Majorana property directly; for the framework value of the effective Majorana mass see [30]. The experiment acts on two levels at once:

- particle physics: whether the neutrino is a Majorana fermion;
- structure: whether the finite Weyl conjugation is realized by the unitary F.

A positive result tests the physical realization of the Fourier symmetry in the dilation data; the detailed-balance ratio (49) and the relaxation rates of §4.2–§4.3 depend jointly on the cross-shell interface and the environment coupling. A negative result refutes the correspondence between the Majorana identification and the Fourier eigenphase, but does not by itself exclude every possible thermalization mechanism.

6.2 The mid-temperature cooling correction

The late cooling laws are identical; the signal that genuinely separates the two microscopic couplings sits in the mid-temperature correction. The factors given in §4.5 are $(1 - 2^{-\beta})^2$ and $(1 - 2^{-\beta})$, differing by about 3% at $\beta = 5$. A controlled realization measures the curve of $d\beta/dt$ against β in the mid-temperature segment and fits the power of the correction factor: power two corresponds to the constant rate, i.e. the shell shift V with unit nonzero matrix elements (§2.1); power one corresponds to the bosonic rate with per-quantum independent coupling, whose appearance would indicate that the dilation also carries per-quantum excitation labels. The Majorana phase tests the Weyl realization; the mid-temperature curve tests the microscopic form of the dissipative coupling.

6.3 Test quantities for controlled realizations

A single equilibrium ratio can be replicated by tuning transition rates by hand and cannot identify the mechanism on its own. Detailed balance, the spectral gap, the relaxation band, the character damping, and the generation-label slow mode can be measured jointly in a controlled device:

Test quantity	Closed form ($p = 7$)	Property
Detailed-balance ratio	$1/7$	bath temperature + branching geometry
Spectral gap (infinite depth)	$0.33856\,\kappa$	Alon–Boppana bound
Spectral gap (chain length n)	$[1 - \frac{2\sqrt{7}}{8}\cos\frac{\pi}{n}]\kappa$; n=7: $0.4041\,\kappa$	finite-chain closed form
Relaxation band	$[0.339,\ 1.661]\,\kappa$, oscillations quantized in $\log 7$	dispersion relation
Unit-shell channel damping	$\kappa/8$, conductor-7^K degenerate	fiber annihilator
Conductor tower spectrum ($K = 2$)	$(9 \pm \sqrt{77})\,\kappa/16 = \{1.1109,\ 0.0141\}\,\kappa$	conductor layering (70)
Generation-label mixing	$\Theta(7^{-(K-1)})\,\kappa$ ($\beta = 1$)	effective superselection
T-dependence of superselection time scale	$\mathrm{gap}_{\mathrm{sel}}/\nu_\beta = (1-7^{-\beta})^2\, 7^{-\beta(K-1)}$	activation energy $(K-1)\log 7$ w.r.t. ν_β
Gap–temperature curve	$1 - 2 \cdot 7^{\beta/2}/(1 + 7^\beta)$	analytic
Joint stationary state	single β, all Boltzmann ratios co-present	no common resonance
Late cooling law	$\beta(t)$ linear, $T \propto 1/t$, rate $\gamma_2/\log 2$	model-independent
Mid-temperature correction	$(1 - 2^{-\beta})^2$ vs $(1 - 2^{-\beta})$	coupling-type criterion (§6.2)
Linear damping near vacuum	channel-independent; dependence only via occupations	mean-field result

In the table, the infinite-depth gap, the chain-length gap, and the β-dependent rows are analytic or closed-form expressions; the generation-label mixing row is a numerical eigenvalue. The Arrhenius row is referred to the unit-shell damping ν_β; at fixed κ, $\mathrm{gap}_{\mathrm{sel}} \propto 7^{-\beta K}$, so fitting directly against κ would overcount the activation energy by one rung. Numerical checks: the finite-chain gap sequence (chain length counted as the number of states, terminal shell included; $7, 15, 25, 41$ give $0.4041, 0.3530, 0.3438, 0.3405$) decreases monotonically to the analytic value 0.33856; the full-ring check of the unit-shell channel damping (mod 49, ten measured representatives of conductor-49 characters) gives 0.125000, within 10^{-15} of $\kappa/8$; the slowest nonzero generation-label modes at $K = 2, 3, 4$ are 0.014065, 0.001901, 0.000268, with successive ratios approaching 1/7; the two-direction check of the joint stationary state (frequencies $\log 2$ and $\log 7$) is in §3.4. In a controlled realization, a measured miss falsifies the corresponding component. All numbers of this section can be reproduced with the accompanying scripts `reproduce_17_spectrum.py` and `reproduce_17_cooling.py`. The environment of a controlled realization need not be genuinely open: a spectrally random finite bath inside the weak-coupling golden-rule window already thermalizes the system to the microcanonically self-consistent temperature (numerical check: a 2000-dimensional random-matrix bath gives total variation about 4% between the occupation distribution and the self-consistent Gibbs distribution, with system and bath fitted temperatures agreeing; an equal-frequency few-mode integrable bath produces no thermalization), and all decisive spectral quantities are protected unchanged by the spectral rigidity (72). Openness is needed only for strict irreversibility and permanent records.

7 Conclusion

This paper set out from the shell-conservation theorem: the evaluation algebra and its inner fluctuations preserve every valuation shell, and the dynamics of relaxation to equilibrium is absent from the existing structure. In filling in this layer, no step at the structural level leaves room for choice. Kinematically, under the clauses of the triple data and the normalization conventions, the cross-shell structure has exactly one realization—the thermalization dilation; the Fourier unitary is unique up to a phase, and the minimal unitary dilation is unique up to unitary equivalence on the auxiliary space. Physically, the environment carrier passes the threefold screening by conservation laws, energy, and records, and under the identification of the equilibrium framework only the photon survives. Dynamically, the weak-coupling limit of repeated collisions converges to the unique GKSL generator, with the rate ratio locked by the KMS condition to $p^{-\beta}$. The mechanism is given by the structure: apart from normalization conventions and the overall rate unit, there was never more than one candidate; the microscopic type of the dissipative coupling (constant or bosonic rate) is left to the mid-temperature criterion.

The results the mechanism delivers fall into two classes: direction and rates. The arrow of time becomes a theorem: at fixed temperature the relative entropy is monotone, the mean entropy production and the trajectory-wise fluctuations issue from detailed balance, and the escaping records are the physical content of the refreshing boundary condition; the entire time asymmetry is contained in that boundary condition, so the dividing line between theorem and assumption is exact. The relaxation spectrum, in turn, is immune to the microscopic branch weights: the spectral gap, the characteristic polynomial, and the conductor ladder are completely determined by tree geometry and detailed balance, and the spectral predictions require no knowledge of the branch weights; the generation labels occupy the slowest rung of the ladder, and their Arrhenius temperature dependence is the mechanism's most direct experimental signature.

Direction and rates meet in quasistatic cooling: the system tracks the Gibbs state moving with temperature, and the late cooling law follows.

Under the identification of the equilibrium framework, the Majorana property of the neutrino and the physical realization of the finite Weyl symmetry are two readings of one and the same Gauss sign, so neutrinoless double beta decay directly tests the physical realization of the mechanism. All test quantities, once expressed in units of κ, are closed-form or analytic, with no independent fitting parameter, and the spectral ones are protected by spectral rigidity; the items are listed in §6. Under the single boundary condition of environment refreshing, the equilibrium framework acquires the dynamical layer it previously lacked: relaxation to equilibrium, the direction of time, and the relaxation rates are no longer external inputs but corollaries of the structure of the state space.

A Shell-invariance verification of the evaluation algebra

The conservation relations of §1.3 concern the evaluation algebra of §1.2. The table below verifies the shell invariance of its generators item by item; the component names and constructions are taken from the equilibrium framework[30], and this paper only verifies their shell behavior.

The convolution operator of the equilibrium framework is $(\hat{K}f)(x) = \sum_{n\geq 1,\, p\nmid n} n^{-\beta} f(nx)$ ($\beta > 1$; the clause $p \nmid n$ guarantees that $x \mapsto nx$ is invertible). For $p \nmid n$ and arbitrary x one has $v(nx) = v(x)$, so $\hat{K}$ preserves the support of every shell and commutes with all shell projections P_j. Its full spectral decomposition proceeds shell by shell (on the terminal shell $S_K = \{0\}$, $\hat{K}$ is multiplication by the constant $(1 - p^{-\beta})\zeta(\beta)$): on the shell $S_j \cong (\mathbb{Z}/p^{K-j}\mathbb{Z})^\times$ the eigenbasis consists of the lifts of multiplicative characters mod p^{K-j}, with eigenvalues the corresponding $L^{(p)}(\beta, \chi)$ (for $\beta \leq 1$ by eigenvalue-wise analytic continuation; the L-values of nontrivial χ are entire, while the ζ factor of the trivial character has a pole at $\beta = 1$, which the continuation excludes).

The components of the framework's evaluation matrix elements and their shell-preservation grounds are:

#	Component	Shell-preservation ground	Verdict
1	phase operators ε_r (position functions)	position-diagonal	preserves
2	unit scalings M_u	$v(ux) = v(x)$	preserves
3	involution ι, complex conjugation	$v(-x) = v(x)$	preserves
4	character projections	intra-shell unit-group directions, per shell	preserves
5	convolution $\hat{K}$, $\hat{K}^{1/2}$	$p \nmid n$ fixes v; $\hat{K}^{1/2}$ by shell-wise functional calculus	preserves
6	insertion (dressing) operators	shell-wise character blocks (from [30])	preserves
7	conditional expectation	projection onto shell-preserving subalgebra	preserves
8	generational readout / lattice projection	intra-shell unit-group diagonal readout ([30])	preserves
9	finite Dirac-type operator	unit-group direction, block-diagonal (input, §1.2)	preserves
10	bare isometries μ_n ($p \mid n$)	$v(n\alpha) = \min(v(n) + v(\alpha), K)$: crosses shells	excluded

Row 10 corresponds to the exclusion clause of §1.2. The multiplicative translations inside evaluation matrix elements pass through $\hat{K}$, whose summation condition contains $p \nmid n$; the p-direction is normalized in §2.1 into the shell shift V and receives its joint unitary realization in §2.2.

If the $p \mid n$ terms are kept, the full multiplicative convolution operator is

$$(\hat{K}_{\text{full}} f)(x) = \sum_{n\geq 1} n^{-\beta} f(nx). \tag{91}$$

The equilibrium framework does not adopt this version. Direct computation exposes its shell-layer structure: for the mod-p^{K-j} character lift $\tilde{\chi}_j$ supported on S_j,

$$\hat{K}_{\text{full}}\, \tilde{\chi}_j = L^{(p)}(\beta,\chi) \sum_{j' \le j} p^{-(j-j')\beta}\, \tilde{\chi}_{j'}^{(\chi)},$$

where $\tilde{\chi}_{j'}^{(\chi)}$ is the (in general imprimitive) lift of χ pulled back to the shell $S_{j'}$ through the mod-p^{K-j} projection; for inputs with $j < K$ this is an exact identity, and the zero-element (terminal-shell) column carries the extra coefficients $\zeta(\beta)p^{-(K-j')\beta}$. The cross-shell components are strictly lower-triangular, each shell of descent carrying a weight $p^{-\beta}$—the same numerical shape as the rate ratio (54) of the weak-coupling limit. The half of the multiplicative motion that the equilibrium framework cut away by the invertibility clause in the definition of its convolution operator is precisely the dissipative channel supplied by this paper; the cut surface retains an isomorphic imprint of the weights.

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